Analytical workflows project: parasite metabarcoding

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My previous experience

- Rstudio user since 2020
  - Analysis of relatively small datasets
  - Made a github account but didn't know how to use it
- CQLS cluster user since 2023
  - Took the linux/unix and biological data analysis classes
  - Handling intense operations on metabarcoding results
  - Utilizing gitlab, python and r scripts
- Learned everything piecemeal
- Why I'm in this class
  - Creating a single cohesive workflow
  - Learning and following "best practices" for now and the future

Metabarcoding is useful for describing parasite communities

- Metabarcoding is a type of DNA sequencing that focuses on describing diversity across a small “barcode”
  - Sequences are converted to a list of organisms detected
  - End result is community composition data for each sample
- Parasites play critical roles in ecosystems
  - Host population dynamics
  - Trophic chains and energy flow
- Parasites are small, cryptic, and hard to identify morphologically, but genetic methods get around that
- My overarching goals are to document the diversity of parasites, host associations, and their relationships with the environment

I need to process my metadata better

- I have fish capture data from 10+ field trips
  - Lengths, weights, types of samples collected
  - Often receiving untranscribed data sheets and handfuls of samples from different people
- Sample metadata
  - When the sample was extracted, sequenced, by who
  - Phyloseq object in R ties metadata to the community composition in one cohesive object
- Multiple metadata files are getting chaotic
- End goal is one giant metadata file with all information, that can be updated and subset as needed

Anticipated challenges

- Catching errors in the transcribed datasheets
- Correctly matching columns that might have different names, but the same type of data
- Integrating fish-wise and sample-wise data
- Workflow and final metadata sheet needs to be easy to update